NAVRACHANA UNIVERSITY

Teaching Plan

Computer Networking (CS308)

Credit : 3

Faculty	Name	Contact	Office Hours	Office Location
	Ninad Chandravadan Bhavsar	ninad.bhavsar@nuv.ac.in	9:00 - 5:00	A-307

School	SET

Semester	Spring (Jan to June 2025)			
Lecture time & Weedays	SECTION_3	Thursday	14:00:00-15:00:00	08-01-2025 TO 09-05-2025
		Monday	15:00:00-16:00:00	08-01-2025 TO 09-05-2025
		Tuesday	15:00:00-16:00:00	08-01-2025 TO 09-05-2025
	SECTION_4	Thursday	14:00:00-15:00:00	08-01-2025 TO 09-05-2025
		Tuesday	15:00:00-16:00:00 14:00:00-15:00:00 16:00:00-17:00:00	08-01-2025 TO 09-05-2025 08-01-2025 TO 23-02-2025 24-02-2025 TO 09-05-2025
	SECTION_1	Monday	11:00:00-12:00:00	08-01-2025 TO 09-05-2025
		Thursday	10:00:00-11:00:00 11:00:00-12:00:00	08-01-2025 TO 09-05-2025 20-01-2025 TO 09-05-2025
		Friday	11:00:00-12:00:00	08-01-2025 TO 19-01-2025
	SECTION_2	Thursday	10:00:00-11:00:00 11:00:00-12:00:00	08-01-2025 TO 09-05-2025 20-01-2025 TO 09-05-2025
		Monday	13:00:00-14:00:00	08-01-2025 TO 09-05-2025
		Friday	11:00:00-12:00:00	08-01-2025 TO 19-01-2025

Pre-requisite	
Co-requisite	
Anti-requisite	
Course Description	This course delves into the details of internet and communication between computer systems. It helps students understand networking concepts using the OSI reference model. OSI, abbreviation for Open Systems Interconnection is a conceptual model for studying networks, its components and protocols. It partitions the communication system into 7 abstract layers and studies importance of each layer in detail
Course Abstract	This course introduces the layered architecture for computer networks. The design and operations of layered architecture based computer network is discussed, with detailed discussion of each layer. The popular protocols at each layer are also discussed.
Course Objectives	• To teach basic organization of network and internet through OSI model. • To introduce different hardware components used for networking computers. • Study of protocols used in networking. • Study of networking architectures • Studying physical aspects of networking such as bandwidth allocation, hardware and tools
Course Outcomes	Student will be able to • CO1 Choose organization of networking elements • CO2 Understand working of different protocols for layers of OSI models • CO3 Apply concepts learned in domains like cloud computing and distributed systems. • CO4 Apply the networking concepts • CO5 Undertsand the security aspects
Course Typology	Theory
Course Outline	UNIT 1: Introduction to layered architecture UNIT 2: Physical layer UNIT 3: Data Link Layer UNIT 4: Network Layer UNIT 5: Transport and Application Layer
Pedagogy	Classroom Teaching with BB/PPT/online tools for initial phase and then PBL (Project Based Learning)
Expectations from Students	Student will be able to • CO1 Choose organization of networking elements • CO2 Understand working of different protocols for layers of OSI models • CO3 Apply concepts learned in domains like cloud computing and distributed systems. • CO4 Apply the networking concepts • CO5 Undertsand the security aspects
Assessment / Evaluation	
Project / Assignment Details	60 marks continous evalution: 20 marks quiz1 + 20 marks quiz2 + 20 marks Wireshark based assignment 40 marks end sem evaluation: Project THERE WILL NOT BE A TYPICAL SIT DOWN THEORY EXAM OF 40 MARKS FOR END SEM EVALUATION. INSTEAD, THE PROJECT WILL BE EVALUATED.
Course Material	Computer Networks: Andrew S Tanenbaum 6th edition Data Communications and Networking: Behrouz Forouzan Cisco Knowledge Center/Online Resources
Additional Information	Technical Specs of Network Devices like Routers, Switches, APs will be discussed.

Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
introduction to layered architecture	1	Need and Applications of a computer network	Computer Networks: Andrew S Tanenbaum	Class Room Teaching, Demo of online tools	
	2	Concept and need of layered architecture			
	3	OSI model and TCP/IP model			
	4	Network Topologies			
	5	Layered Architecture Operations			
	6	Peer Layers, Virtual and Real Communications			
	7	End to End and Hop to Hop Communications			
	8	Brief overview of popular protocols at each layer			
	9	Brief Funtionality of each layer			
Physical Layer	10	Basic Functions of physical layer			
	11	Analog and Digital Signals			
	12	Wired Transmission Media			
	13	Wireless Media			
	14	Working of Fiber optic Systems			
	15	Noise, Attenuation, Distortion, Bandwidth, Delay, Jitter			
	16	A/D and D/A conversion			
	17	Modulation Schemes			
	18	Multiplexing Schemes			
Data Link Layer	19	Basic Functions of Data Link layer			
	20	Flow Control, Bit Stuffing, Byte Stuffing			
	21	Error Control: CRC, Hamming code, Parity based Methods			
	22	Simple Protocols			
	23	Sliding Window Protocols: Go back n, selective repeat			
	24	Introduction to MAC			
	25	Aloha, Slotted Aloha, CSMA/CD			
	26	CSMA/CA			
	27	Wired and Wireless LANs			
Network Layer	28	Basic Functions of Network layer			
	29	Routing and Forwarding, basic working of a router			
	30	IPv4 and IPv6			
	31	Routing Protocols			
	32	DV based Protocols			
	33	LSR based Protocols			
	34	Bellman Ford Routing Algorithm, Dijkstra's shortest path			
	35	OSPF, BGP			
	36	IGMP, inter AS and intra AS routing protocols			
Transport and Application Layer	37	Basic Functions of Transport and Application Layer			
	38	TCP			
	39	UDP			
	40	TCP and UDP based applications			
	41	HTTP and HTTPS			
	42	DNS			
	43	SMTP, IMAP, POP3			
	44	FTP, TELNET, SSH			
	45	Socket Programming			

Publish Status	Published